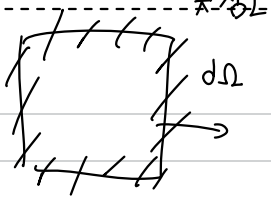
5주차 Last Lecture.

iterative method

$\partial\Omega = \partial\Omega_D \cup \partial\Omega_N$

Dirichlet bc $u = f$

Neumann bc $\frac{\partial u}{\partial n} = f$

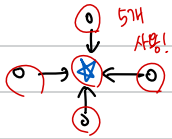


*참고

$$\begin{cases} \Delta u = f & in \ x \in \Omega \\ u = g = 0 & in \ x \in \partial\Omega \end{cases}$$

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f \qquad let \ h = \Delta x = \Delta y$$

u_{ij} : x, y



$$\Rightarrow \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2} = f_{ij}$$

$$\frac{1}{h^2}\left(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}\right) = f_{i,j}$$

인덱싱 계산 $i \times (ny-1) + j$

경계 방식 $\begin{cases} i' = i+1 \\ j' = j+1 \end{cases}$

$\Rightarrow Ax = b$ 꼴로 나타냄. (모든 i, j에 대해 나타내면 $\Leftarrow$ 가능) $Ax=b$

$\Rightarrow$

$$\begin{bmatrix} -4 & 1 & \cdots & 1 & \cdots \\ 1 & -4 & 1 & \cdots & 1 \cdots \\ \vdots & & & & \\ & 1 & 1 & -4 & 1 \cdots \\ & \vdots & & & \\ & & & 1 \cdots 1 & -4 \end{bmatrix} \begin{matrix} \\ \\ \\ \\ \\ \\ \end{matrix} \begin{bmatrix} u_{11} \\ u_{12} \\ \vdots \\ u_{i,j} \\ \vdots \\ u_{(nx-2)(ny-2)} \end{bmatrix} \begin{matrix} \\ \\ \\ \\ \\ \\ \end{matrix} = \begin{bmatrix} f_{11} \\ f_{12} \\ \vdots \\ f_{ij} \\ \vdots \\ f_{(nx-2)(ny-2)} \end{bmatrix} - \frac{1}{h^2} \begin{bmatrix} f_{01} + f_{10} \\ f_{02} \\ \vdots \\ 0 \\ \vdots \\ f_{(nx-1)(ny-2)} \end{bmatrix}$$

$(nx-1)(ny-1)$

$(i'+1, j'+2)$, $(i'+1, j')$, $(i'+1, j'+1)$

A $\quad$ X $\quad$ b

Jacobian, Gauss-Seidel

$Ax=b$

$x^{k+1} = Gx^k + C \quad k=1, \cdots m$ (수렴할 때 까지)

$\|Ax-b\| < \varepsilon$ (residual or norm)

$\|x^{k-1} - x^k\| < \varepsilon'$

2가지 방법 중 ~

$x = A^{-1} \cdot b$

어떻게..?

G, C 어떻게 정의?

- Gauss-Seidel Method.

$A = M - N$

$G = M^{-1}N$

$C = M^{-1}b$

$Ax = b$

$\Rightarrow (M-N)x = b$

$\Rightarrow Mx^{k+1} = Nx^k + b$

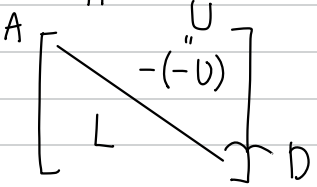
$\Rightarrow x^{k+1} = M^{-1}Nx^k + M^{-1}b$

$A = D + L - U$ (M, N)

$-D$: diag

L : lower tri

U : upper tri



$G = M^{-1}N = (D+L)^{-1}(-U)$

$C = M^{-1}b = (D+L)^{-1}b$

$x^{k+1} = Gx^k + C$

$(D+L)^{-1}x^{k+1} = -Ux^k + b$ ←?

$\Rightarrow Dx^{k+1} = -Lx^{k+1} - Ux^k + b$

$\therefore x^{k+1} = D^{-1}(-Lx^{k+1} - Ux^k + b)$

$x^{k+1} = (D+L)^{-1}(-U)x^k + (D+L)^{-1}b$

$(D+L)x^{k+1} = (-U)x^k + b$ //

양변항 Poisson eq 사용.

x_i

$\Rightarrow x_i^{k+1} = \frac{1}{a_{ii}}\left(-\sum_{j<i} a_{ij}x_i^{k+1} - \sum_{j>i} a_{ij}x_i^k + b_i\right)$??

